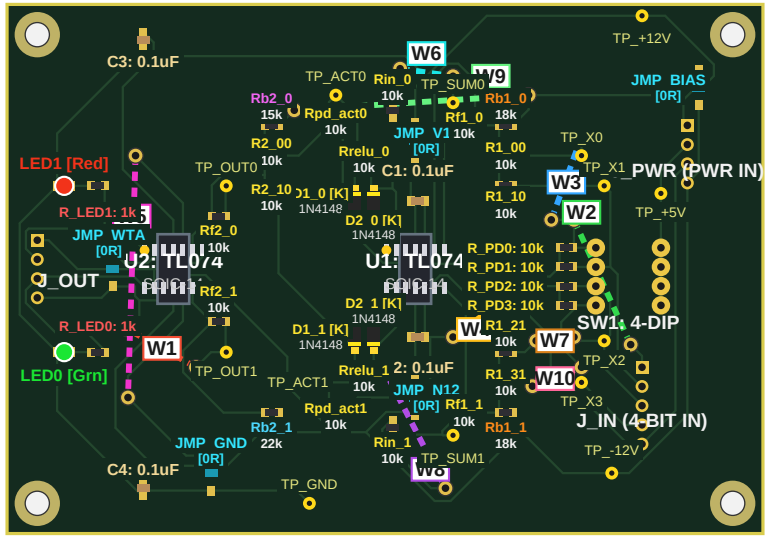
ANALOG NEURAL NETWORK TEST BOARD (70x100mm) - SMD ASSEMBLY MAP

FLIPPED VIEW [X-MIRRORED] - ALL TEXT IN CORRECT UPRIGHT ORIENTATION | SCALE: 1:1 TRUE SIZE

Designed for bottom/translucent inspection & hand-soldering. Traces/Pads are flipped; Text is upright left-to-right.

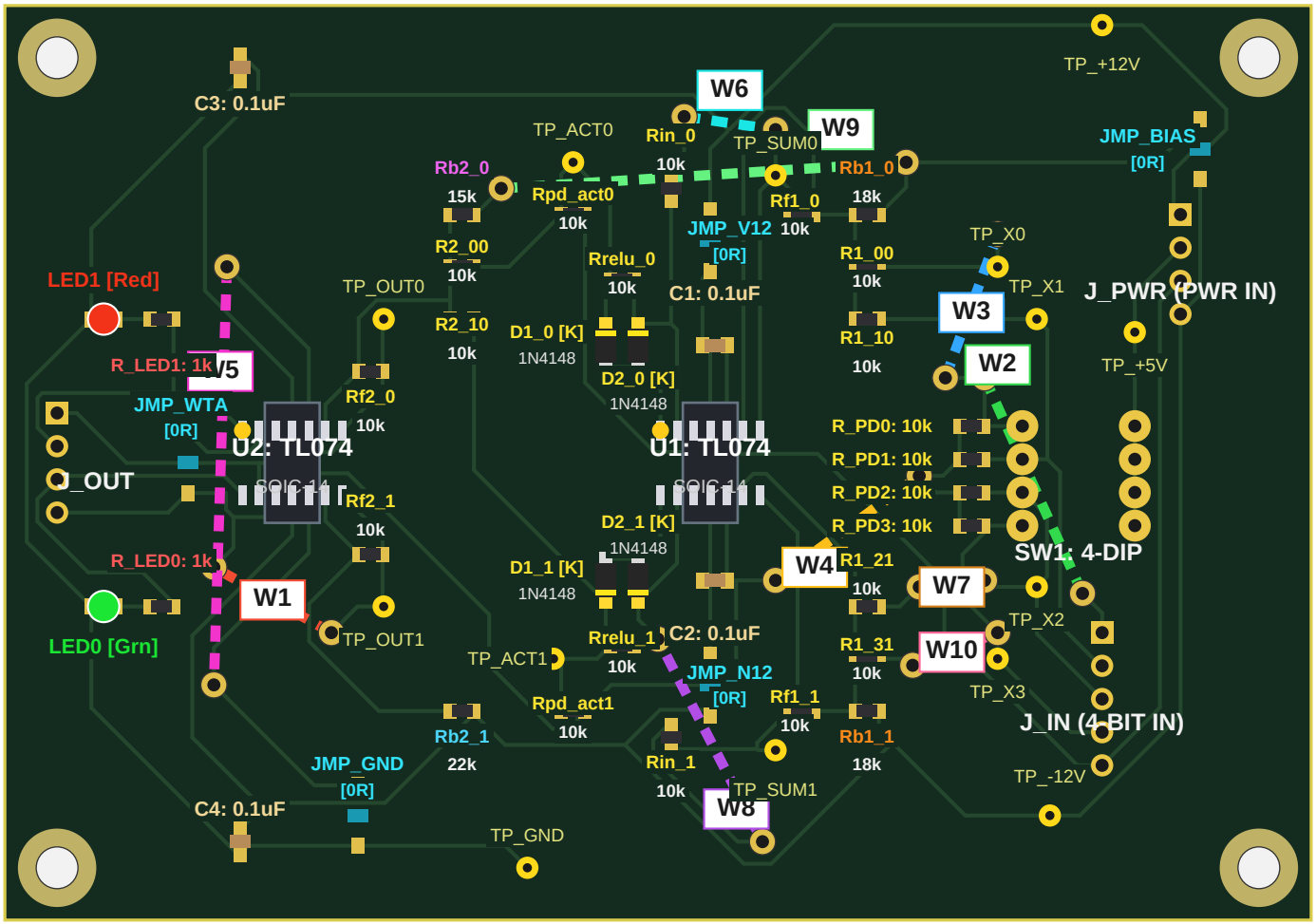


50.0 mm [1:1 TRUE SCALE CALIBRATION BAR - MEASURE WITH CALIPERS]

LEGEND: [Gold] TL074 ICs | [Yellow] 10k Resistors | [Orange] 18k Bias | [Pink] 15k Bias | [Cyan] 22k Bias | [Red] 1k Limiters | [Dashed Lines] W1..W10 Jumpers | [Cyan] 0R Jumpers

ANALOG NEURAL NETWORK TEST BOARD - ENLARGED BENCH SOLDERING GUIDE

FLIPPED VIEW [X-MIRRORED] | ALL TEXT UPRIGHT | 1.82x BENCH SCALE



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ANALOG NEURAL NETWORK TEST BOARD - SMD BILL OF MATERIALS & CHECKLIST

Flipped View Assembly Guide | Component Directory with Resistance Values, Packages, Footprints & Polarity

DONE	DESIGNATOR	VALUE / MODEL	PACKAGE	QTY	FUNCTIONAL BLOCK & NOTES
☐	U1, U2	TL074	SOIC-14	2	Quad Op-Amp: U1=Layer 1 Sum+ReLU, U2=Layer 2 Sum+WTA [Pin 1 Dot]
☐	D1_0, D1_1	1N4148	SOD-123	2	Layer 1 Precision Inverting Half-Wave ReLU Series Diodes [Cathode K]
☐	D2_0, D2_1	1N4148	SOD-123	2	Layer 1 Precision ReLU Feedback Clamp Diodes [Cathode K]
☐	LED1	Red / Vert	0805 SMD	1	Winner-Take-All Vertical Class Winner Indicator [Cathode to GND]
☐	LED0	Green / Horiz	0805 SMD	1	Winner-Take-All Horizontal Class Winner Indicator [Cathode to GND]
☐	R1_00, 10, 21, 31	10k 1%	0805 SMD	4	Neuron Input Summing Resistors (X0..X3 into Hidden Layer 1)
☐	Rf1_0, Rf1_1	10k 1%	0805 SMD	2	Hidden Layer 1 Feedback Resistors (Gain = 1.0)
☐	Rin_0, Rin_1	10k 1%	0805 SMD	2	ReLU Inverter Input Coupling Resistors
☐	Rpd_act0, act1	10k 1%	0805 SMD	2	ReLU Active Output Stage Pull-down to Ground
☐	Rrelu_0, relu_1	10k 1%	0805 SMD	2	ReLU Inverting Translinear Feedback Resistors
☐	R2_00, R2_10	10k 1%	0805 SMD	2	Output Layer 2 Hidden Activation Weight Resistors
☐	Rf2_0, Rf2_1	10k 1%	0805 SMD	2	Output Layer 2 Summing Feedback Resistors
☐	R_PD0..R_PD3	10k 1%	0805 SMD	4	DIP Switch 4-Channel Input Pull-Down Resistors (X0..X3 to GND)
☐	Rb1_0, Rb1_1	18k 1%	0805 SMD	2	Hidden Layer 1 Negative Bias Resistors (Connected to -12V Rail)
☐	Rb2_0	15k 1%	0805 SMD	1	Output Class 0 Bias Resistor (Connected to -12V Rail)
☐	Rb2_1	22k 1%	0805 SMD	1	Output Class 1 Bias Resistor (Connected to -12V Rail)
☐	R_LED0, R_LED1	1k 5%	0805 SMD	2	LED Current Limiting Resistors (From Comparator Outputs)
☐	C1, C2, C3, C4	0.1uF 50V	0805 SMD	4	Power Supply Decoupling MLCC Ceramic Caps (+12V & -12V Rails)
☐	JMP_BIAS, V12, N12	0 Ohm (0R)	0805 SMD	3	Power Rail Planar Corridors (-12V Bias, +12V Rail, -12V Rail)
☐	JMP_WTA, JMP_GND	0 Ohm (0R)	0805 SMD	2	WTA Comparator Cross-Link Corridor & Ground Plane Return
☐	W1..W10	Wire Jumpers	1.0mm TH	10	Through-Hole Insulated Wire Jumper Bridges (See Flipped Wire Routing Map)
☐	SW1	4-DIP Switch	SMD Low-Prof	1	Manual 4-Bit Binary Pattern Input Selector (X0..X3)
☐	J_PWR	PinHeader 1x04	2.54mm TH	1	Main Power Connector: Pin 1=+12V, 2=+5V, 3=GND, 4=-12V
☐	J_IN	PinHeader 1x05	2.54mm TH	1	External Signal Input Connector: Pins 1..4=X0..X3, Pin 5=GND
☐	J_OUT	PinHeader 1x04	2.54mm TH	1	WTA Output Header: Pin 1=OUT0, 2=OUT1, 3=+5V, 4=GND

RECOMMENDED HAND-SOLDERING ORDER FOR BEST ACCESSIBILITY:

1. Solder SOIC-14 ICs (U1, U2) first -> 2. Diodes & 0805 Resistors/Caps -> 3. LEDs & 0-Ohm Jumpers -> 4. Wire Jumpers W1..W10 -> 5. Headers & Switch